

Mathematical Methods for Differential Co-expression and QUBO Formulation

1 Introduction and Objective

The objective is to find, for a given biological pathway, the most effective gene regulatory mechanisms. This is achieved by identifying a subnetwork of core pathway genes that exhibits the maximum differential regulatory change in co-expression between Knockout (**KO**) and Wild Type (**WT**) conditions. This approach aims to uncover biologically meaningful core gene regulatory mechanisms within the pathway.

This problem is formulated for a quantum annealing solution using the **Quadratic Unconstrained Binary Optimization (QUBO)** framework. The optimization goal is to minimize an energy function $E(\mathbf{z})$ over a binary vector $\mathbf{z} \in \{0, 1\}^G$, where $z_i = 1$ indicates gene i is selected for the regulatory subnetwork.

2 Gene Expression Data Preprocessing

The input data is the normalized single-cell RNA sequencing count matrix $\mathbf{X}_{\text{sub}} \in \mathbb{R}^{G \times N}$, where G is the number of genes and N is the number of cells. All elements of $\mathbf{X}_{\text{sub}}$ are non-negative, $x_{i,j} \geq 0$.

2.1 Condition-Specific Subsetting

The matrix is split into condition-specific matrices:

$$\begin{aligned}\mathbf{X}_{\text{ko}} &= \mathbf{X}_{\text{sub}}(:, \text{idx}_{\text{KO}}) \in \mathbb{R}^{G \times N_{\text{KO}}} \\ \mathbf{X}_{\text{wt}} &= \mathbf{X}_{\text{sub}}(:, \text{idx}_{\text{WT}}) \in \mathbb{R}^{G \times N_{\text{WT}}}\end{aligned}$$

3 Non-Negative Gene-to-Gene Cosine Similarity

The similarity matrix for each condition is calculated as the Gram Matrix of the row-normalized gene expression vectors, representing a non-negative Cosine Similarity. This non-negative normalization approach is chosen deliberately: by maintaining positive values, it allows for a clearer assessment and easier tracking of gene regulatory changes across the two conditions, whereas common standardization (Z-scoring) could obscure the original directionality insights.

3.1 Row Normalization (L_2)

Each gene's expression vector (row $\mathbf{x}_g$) is normalized to a unit vector $\mathbf{z}_g$ using the L_2 norm:

$$\mathbf{z}_g = \mathbf{X}_{\text{norm}}(g, :) = \frac{\mathbf{x}_g}{\|\mathbf{x}_g\|_2}, \quad \text{where } \|\mathbf{z}_g\|_2 = 1$$

This step yields the normalized matrices $\mathbf{X}_{\text{ko_norm}}$ and $\mathbf{X}_{\text{wt_norm}}$.

3.2 Similarity Matrix ($\mathbf{X}_{\text{norm2}}$)

The similarity matrix is the dot product of the normalized matrix and its transpose. Since all elements in $\mathbf{X}_{\text{norm}}$ are non-negative, the resulting similarity $\mathbf{X}_{\text{norm2}}(i, j)$ is bounded in $[0, 1]$, quantifying the non-negative Cosine Similarity between gene i and gene j .

$$\begin{aligned}\mathbf{X}_{\text{ko2norm}} &= \mathbf{X}_{\text{ko_norm}} \cdot \mathbf{X}_{\text{ko_norm}}^T \\ \mathbf{X}_{\text{wt2norm}} &= \mathbf{X}_{\text{wt_norm}} \cdot \mathbf{X}_{\text{wt_norm}}^T\end{aligned}$$

4 Differential Co-expression Matrix

The **Differential Co-expression Matrix** $\mathbf{X}_{\text{diff}}$ is the difference between the two similarity matrices.

$$\mathbf{X}_{\text{diff}} = \mathbf{X}_{\text{wt2norm}} - \mathbf{X}_{\text{ko2norm}}$$

A key property of this differential matrix is that since both $\mathbf{X}_{\text{wt2norm}}$ and $\mathbf{X}_{\text{ko2norm}}$ are similarity matrices with unit diagonals by construction, the diagonal elements of the difference matrix are naturally zero, $\mathbf{X}_{\text{diff}}(i, i) = 1 - 1 = 0$.

A negative value $\mathbf{X}_{\text{diff}}(i, j) < 0$ indicates increased co-expression for gene pair (i, j) in the **KO** condition relative to **WT**.

5 QUBO Formulation

The objective is to find a subnetwork of size K that maximizes the co-expression increase in the **KO** condition.

5.1 Initial QUBO Matrix ($\mathbf{Q}_1$)

The base matrix $\mathbf{Q}_1$ combines several terms influencing the gene selection:

$$\mathbf{Q}_1 = \mathbf{X}_{\text{diff}} + \text{diag}(\mathbf{V}_{\text{diff}}) - (\mathbf{X}_{\text{wt_mnn}} + \mathbf{X}_{\text{ko_mnn}}) + \mathbf{X}_{\text{net_target}}$$

where:

- $\mathbf{V}_{\text{diff}}$ is the gene-wise difference in cell state projection (linear term).
- $\mathbf{X}_{\text{wt_mnn}} + \mathbf{X}_{\text{ko_mnn}}$ incorporates Mutual Nearest Neighbor (MNN) information, representing pairs of genes that are very likely to interact due to high local similarity. This term is subtracted to favor the selection of highly connected/interacting genes.
- $\mathbf{X}_{\text{net_target}}$ is a term introduced to ensure the final QUBO solution is highly focused on the declared pathway genes (e.g., Wnt pathway genes).

5.2 Optimization Goal (Quadratic Term)

Quantum annealing is a method that minimizes the energy function $E(\mathbf{z})$ (the cost function). To achieve the biological objective of *maximizing* the differential co-expression increase in **KO** (i.e., maximizing the negative values in $\mathbf{X}_{\text{diff}}$), the quadratic term is set to the negative of the desired objective:

$$Q_{i,j}^{\text{objective}} \propto -\mathbf{X}_{\text{diff}}(i, j)$$

Thus, minimizing the energy $E(\mathbf{z})$ is equivalent to maximizing the beneficial (negative) components of the differential co-expression.

5.3 Subnetwork Size Constraint

The primary role of this constraint is to enforce that the optimal solution $\mathbf{z}$ selects exactly K features from the gene subnetwork. A penalty term is added to the objective function:

$$E_{\text{constraint}} = A \left(\sum_{i=1}^G z_i - K \right)^2$$

where A is a large penalty scalar, typically $A \approx 10 \cdot \max(|\mathbf{Q}_1|)$.

5.4 Final QUBO Matrix ($\mathbf{Q}$)

The final QUBO matrix $\mathbf{Q}$ is constructed by incorporating the penalty A into the elements of $\mathbf{Q}_1$.

1. **Off-Diagonal Terms ($i \neq j$):** The penalty A is added to all off-diagonal elements. This penalty ensures that the linear terms (diagonal elements) derived from the size constraint dominate the optimization landscape when the constraint is violated, effectively increasing the overall cost function values across the matrix to bias the solution toward satisfying K .

$$Q_{i,j} = \mathbf{Q}_1(i, j) + A \quad \text{for } i \neq j$$

2. **Diagonal Terms ($i = j$):** The diagonal terms of the penalty $E_{\text{constraint}}$ are $A(1-2K)z_i + \text{constant}$. This yields the final diagonal term, which is essential for minimizing the cost when exactly K genes are selected:

$$Q_{i,i} = \mathbf{Q}_1(i, i) + A(1 - 2K)$$

The QUBO problem is then defined by the matrix $\mathbf{Q}$:

$$\text{Minimize } E(\mathbf{z}) = \mathbf{z}^T \mathbf{Q} \mathbf{z} = \sum_{i=1}^G \sum_{j=i}^G Q_{i,j} z_i z_j$$